# Project: Aura Retail OS

## Subtask 1: System Design Proposal

## Course: Object Oriented Programming (IT620)

## Guided By: Prof. Sourish Dasgupta & TAs


**Group 15**
**Group Name: Ghost Compilers**

| Student ID | Name |
|---|---|
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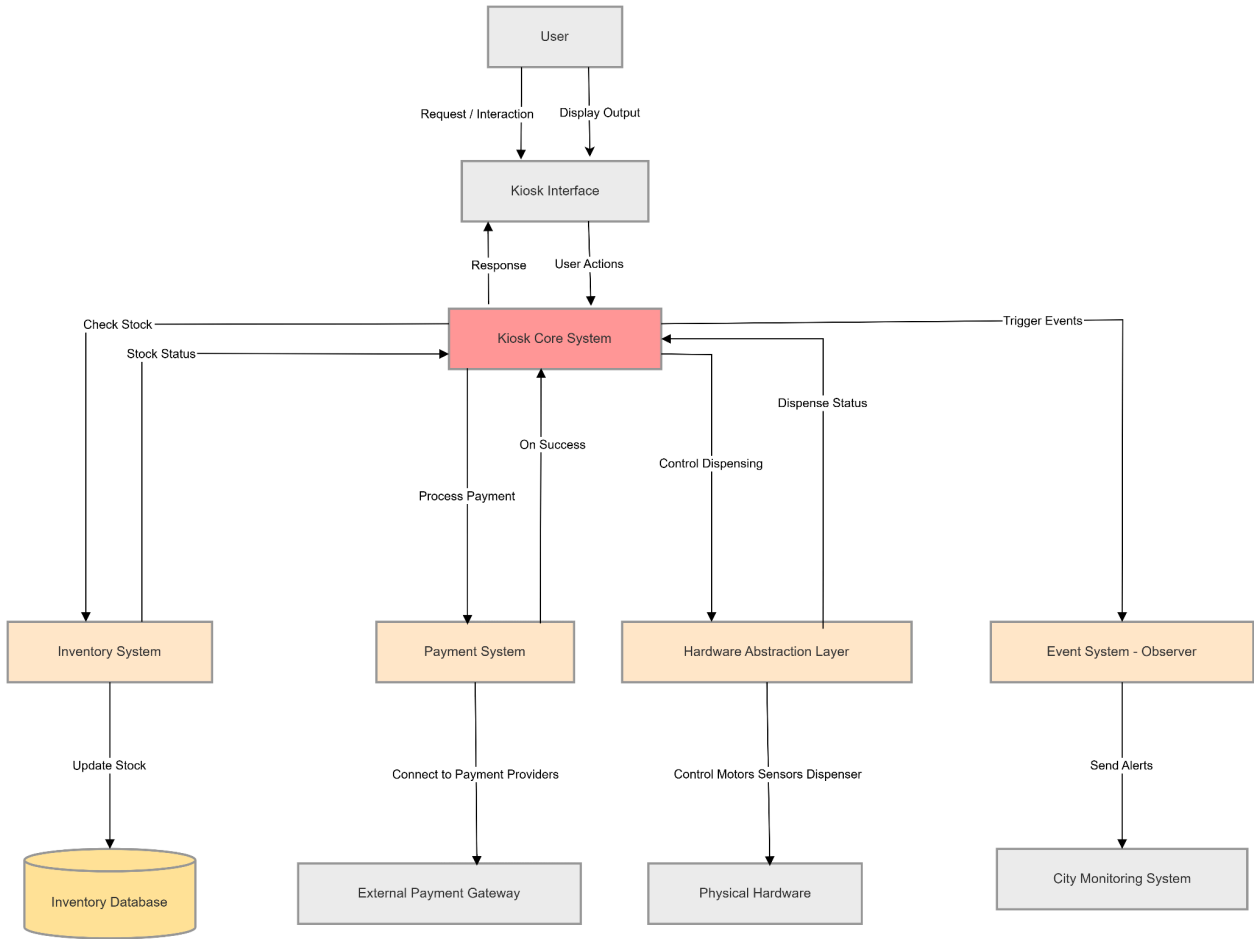
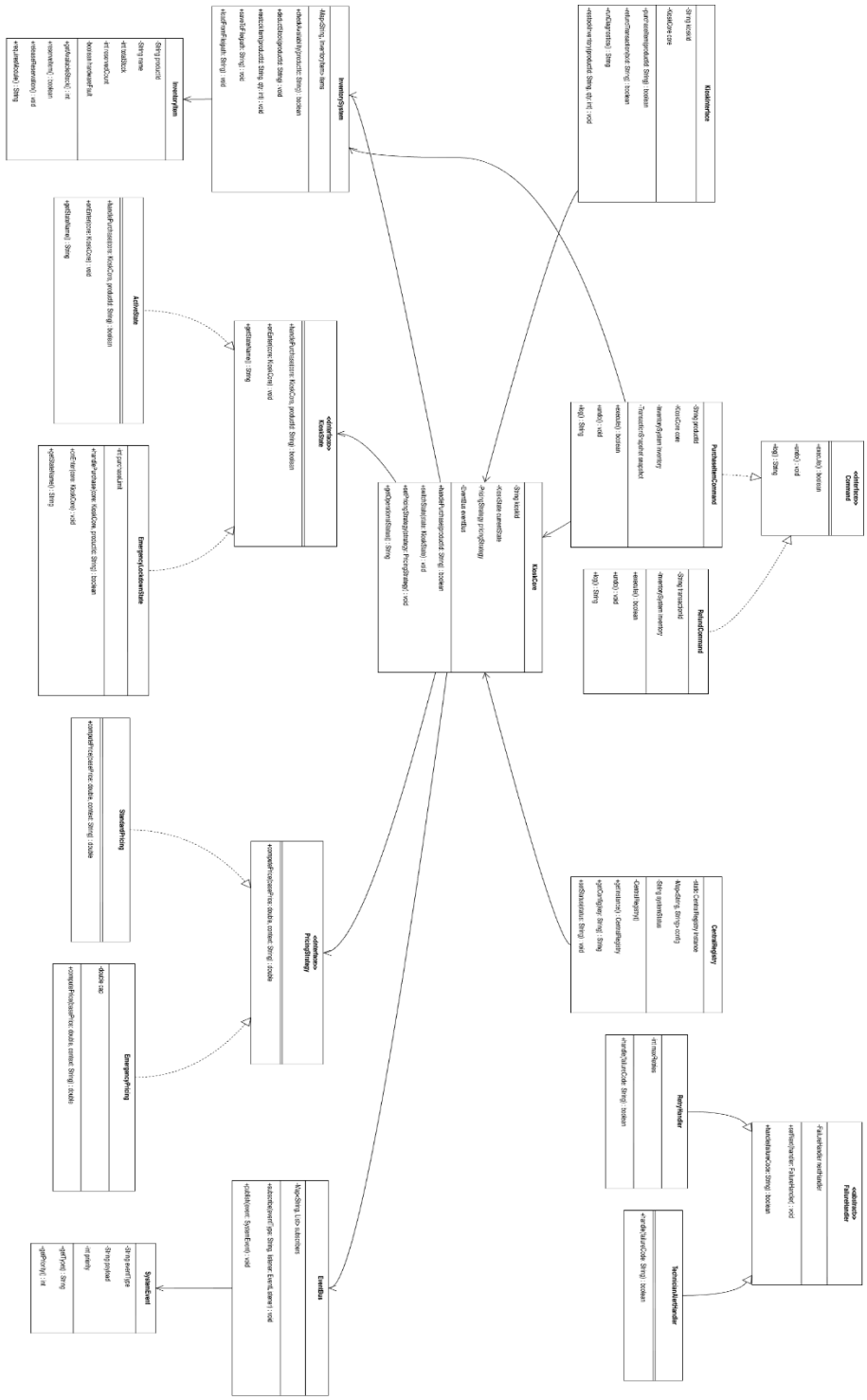
# 1) High-level architecture diagram




Link for Diagram:
https://drive.google.com/file/d/15IivRJV5_EwNGHFisxT193I2J7A7TZXs/view?usp=sharing

## 2) Preliminary class diagram




Link for diagram: [https://drive.google.com/file/d/1SZtvSVGZJoa2hxEcwcRW6fWnviEjYWRY/view?usp=sharing](https://drive.google.com/file/d/1SZtvSVGZJoa2hxEcwcRW6fWnviEjYWRY/view?usp=sharing)

## 3) Planned design patterns

| No | Pattern | Where Used | Why Used |
|---|---|---|---|
| 1 | **Singleton** | CentralRegistry | Ensures one global instance for system configuration and status |
| 2 | **Facade** | KioskInterface | Provides a single clean entry point for all kiosk operations |
| 3 | **Abstract Factory** | Kiosk Creation | Creates compatible component families for each kiosk type |
| 4 | **Command** | Transaction System | Wraps operations into objects for execution, logging, and rollback |
| 5 | **Observer** | Event Bus | Notifies subsystems through events without direct dependencies |
| 6 | **Strategy** | Pricing System | Allows pricing rules to be swapped dynamically at runtime |
| 7 | **State** | Kiosk Modes | Changes kiosk behaviour based on its current operational mode |
| 8 | **Chain of Responsibility** | Failure Handler | Escalates hardware failures step by step before human alert |
| 9 | **Memento** | Transaction Rollback | Saves and restores state to guarantee atomic transactions |

## 4) Subsystem responsibilities of each member

**Hitarth Shah** — Kiosk Core System
Subsystem: Kiosk Core + Central Registry

**Design Patterns:**
Singleton (CentralRegistry)
State (Kiosk Modes — Active, Emergency, Maintenance, Power-Saving)
Facade (KioskInterface)

**Deliverables:**
Main KioskInterface class
CentralRegistry implementation
State classes for all 4 modes
System Architecture Diagram (overall structure)
Leads the Final Presentation

**Nirmal Patel**— Hardware Abstraction Layer
Subsystem: Hardware Layer + Failure Handling

**Design Patterns:**
Chain of Responsibility (Retry → Recalibration → Technician Alert)
Memento (Transaction Rollback / Snapshot before dispensing)
Abstraction (Dispenser, Scanner, Sensor interfaces)

**Deliverables:**
Hardware interfaces and concrete classes
FailureHandler chain classes
TransactionMemento + KioskCaretaker classes
Sequence Diagram (purchase flow)

**Mahi Kansara**— Inventory System
Subsystem: Inventory + Kiosk Factory

**Design Patterns:**
Abstract Factory (PharmacyKioskFactory, FoodKioskFactory, EmergencyKioskFactory)
Composite (Product + ProductBundle)

**Deliverables:**
All inventory classes (Product, Bundle, InventoryManager)
All factory classes and interfaces
JSON/CSV persistence for inventory
Class Diagram (full system)

**Jay Trivedi** — Payment + Event System
Subsystem: Payment System + City Monitoring

**Design Patterns:**
Strategy (StandardPricing, DiscountPricing, EmergencyPricing)
Command (PurchaseItemCommand, RefundCommand, RestockCommand)
Observer (EventBus, LowStockEvent, HardwareFailureEvent, subscribers)

**Deliverables:**
All payment and pricing strategy classes
All Command classes with execute/undo
EventBus + all event/subscriber classes
System Design Report + README